SOFTWARE TEST PLAN (STP)

Customer Relationship Management System

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Document Control

Field	Details
Project	Customer Relationship Management (CRM)
Document	Software Test Plan
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Scope baseline	GitHub Project Management issues visible in the supplied screenshots (#6–#34)
Status	Revised Draft — team review and instructor approval pending

Approvals

Role	Name
Team Member	Shweta
Team Member	Avaneesh N
Team Member	Vishwas V Angadi
Team Member	Krishna Pai
Instructor / Reviewer	To be completed

1. Introduction

1.1 Purpose

This document defines the testing scope, strategy, environment, schedule, roles, risks, test cases and traceability for the CRM application. It follows the supplied Software Test Plan template and uses the issue-aligned SRS requirement identifiers.

1.2 Scope

Testing covers authentication and user management; lead/customer management; customer engagement and support; sales/campaigns; dashboards/reports; audit logging; GitHub Actions/PR checks; system/regression testing; and the performance, persistence, modularity, safe-error and access-protection requirements.

1.3 References and Definitions

- GitHub CRM Project Management board and repository.
- FR: Functional Requirement; NFR: Non-Functional Requirement; RBAC: Role-Based Access Control; RTM: Requirements Traceability Matrix.

2. Test Items

- Login/logout, authentication and password validation.
- Role-based access control and user account management.

- Lead creation/management, lead conversion, customer profiles and customer assignment.
- Customer interaction tracking and follow-up management.
- Customer feedback and retention management.
- Sales and campaign management.
- Dashboards, reports and audit logging.
- GitHub Actions and pull-request checks.
- System/regression testing and project documentation.
- NFR checks: dashboard/list performance, persistent storage, modularity, safe error handling, secret protection and protected pages/APIs.

3. Features to be Tested

SRS Requirement IDs	Feature / Module	Related Project Issue
CRM-F-001 to CRM-F-004; CRM-SR-001 to SR-003	Authentication, password validation, RBAC and user accounts.	#6–#10
CRM-F-005 to CRM-F-008	Lead management, conversion, customer profiles and assignment.	#11–#15
CRM-F-009 to CRM-F-012	Interactions, follow-ups, feedback and retention.	#16–#19, #21
CRM-F-013 to CRM-F-016	Sales, campaigns, dashboards/reports and audit logging.	#22–#25
CRM-F-017 to CRM-F-019	GitHub Actions/PR checks, system/regression testing and documentation.	#26–#28
CRM-NF-001 to CRM-NF-005; CRM-SR-004 to SR-005	Performance, persistence, modularity, safe errors, secret protection and access protection.	#29–#32, #34

4. Features Not to be Tested

- Features or integrations not represented in the project issue list or implemented CRM scope.
- External payment gateway settlement or bank integrations not part of the project.
- Mobile applications, third-party ERP integrations or production infrastructure not specified in the project scope.
- Performance targets beyond the agreed development/test environment measurements.

5. Test Approach / Strategy

5.1 Test Levels and Types

Level / Type	Approach
Functional testing	Verify each implemented CRM feature against its SRS requirement.
Integration testing	Verify frontend-to-backend communication and backend-to-database operations.
System testing	Execute end-to-end workflows across authentication, CRM activities, sales and reporting.
Regression testing	Repeat core workflows after changes to verify previously working features remain functional.
Security testing	Verify authentication, RBAC, protected pages/APIs, safe errors and secret protection.
Performance testing	Measure dashboard and list loading under documented test conditions.
Persistence testing	Verify saved records remain after refresh and application restart.
Maintainability review	Review modularity and separation of responsibilities.
CI / PR verification	Verify configured GitHub Actions and pull-request checks.

5.1.1 Security Validation

- Verify valid and invalid login behavior and logout.

- Attempt to access protected pages and APIs without authentication; access must be rejected.
- Verify role-specific permitted and forbidden operations through the backend API.
- Verify that unauthorized users cannot modify protected CRM records.
- Submit invalid requests and verify safe error responses.
- Inspect responses and available logs to ensure passwords, tokens and secrets are not disclosed.

5.2 Entry Criteria

- A stable project build is available.
- Frontend, backend and database are configured and running.
- Test accounts for implemented roles are available.
- The issue-aligned SRS and SAD are available.
- The configured CI workflow is accessible for verification.

5.3 Exit Criteria

- All planned test cases have an execution status recorded.
- Core functional and security tests have been executed.
- Critical defects are resolved or documented.
- Regression tests have been completed after fixes.
- The RTM and test summary reflect actual evidence; no unexecuted test is marked as passed.

6. Test Environment

Category	Environment / Tools
Hardware	Team development laptops/desktops.
Frontend	Project's configured React/TypeScript frontend.
Backend	Project's configured Node.js/Express backend.
Database	Configured project database and data/model layer.
Tools	VS Code, browser developer tools, GitHub Actions, database inspection and API testing tools.
Test data	Dummy staff accounts, leads, customers and associated CRM records.

7. Test Schedule

Activity	Timing	Reference
Test case review	Before execution	SRS / STP review
Environment smoke test	Before functional testing	Build and database availability
Functional testing	During final verification	Issues #10, #15, #21
Security and NFR testing	During final verification	Issues #29–#34 and security requirements
Regression testing	After fixes / before final demo	Issue #27
Documentation review	Before submission	Issue #28

8. Test Deliverables

- Software Test Plan.
- Test case execution records and evidence.
- Defect reports and retest results, where applicable.

- Updated Requirements Traceability Matrix.
- Final Test Summary Report.

9. Roles and Responsibilities

Team Member	Responsibility
Shweta	Project coordination and contributions to sales, analytics, testing and documentation work.
Avaneesh N	Development contribution to lead and customer management work.
Vishwas V Angadi	Team member; detailed assignment to be confirmed from repository history.
Krishna Pai	Team member; detailed assignment to be confirmed from repository history.

GitHub screenshots show account handles for issue assignees. This document does not infer a real-name mapping for handles where it has not been confirmed.

10. Risks and Mitigation

Risk	Mitigation
Unstable build	Run smoke tests before full execution.
Database unavailable	Verify service availability and connection before testing.
Incorrect role restrictions	Test direct API access for each role, including forbidden actions.
Performance varies by machine	Record machine, dataset and conditions with measured timings.
Regression after fixes	Repeat core authentication, lead/customer, engagement, sales and report tests.
CI workflow does not trigger	Review workflow configuration and pull-request check results.

11. Assumptions and Dependencies

- The supplied GitHub issue screenshots represent the agreed project scope baseline.
- The final application build and configured database are available.
- The team can use dummy test records and role-specific accounts.
- GitHub Actions permissions and workflows are configured in the repository.

12. Suspension and Resumption Criteria

Testing may be suspended if the application cannot start, the database is unavailable, authentication prevents all test users from proceeding, or a blocking defect prevents execution of most planned tests. Testing resumes after the environment is restored, blocking defects are addressed and required test data is available.

13. Test Case Management and Traceability

Execute each test and record the actual result, date, tester and evidence. Status values are Not Run, Pass, Fail or Blocked. The workflow below shows the intended execution path; a failed test is logged, fixed and retested before closure.

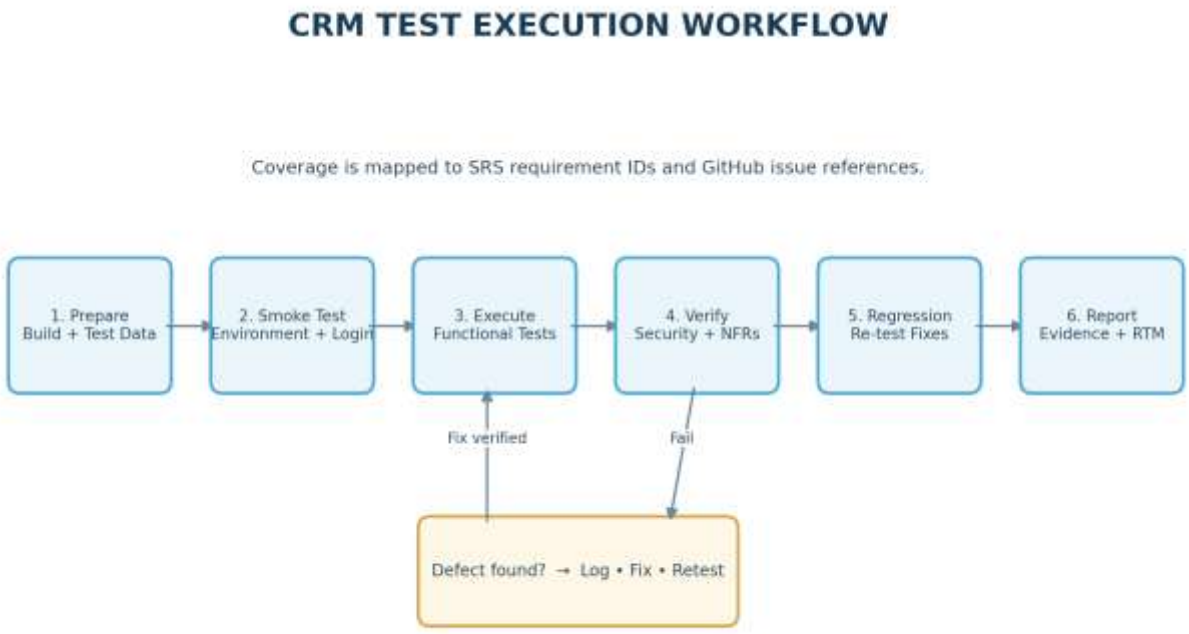


Figure 1. CRM test execution workflow and defect retest loop.

Test ID	SRS Requirement	Module	Test Objective / Procedure Summary
TC-01	CRM-F-001, F-002	Authentication	Verify valid login, invalid login, password validation and logout.
TC-02	CRM-SR-001, NF-005	Protected access	Verify unauthenticated access to protected pages and APIs is rejected.
TC-03	CRM-F-003, SR-002	Role-based access	Verify each role can perform permitted operations and forbidden operations are denied.
TC-04	CRM-F-004	User accounts	Verify authorized user account management operations are saved correctly.
TC-05	CRM-F-005	Lead management	Create, view and update a lead; verify the saved values.
TC-06	CRM-F-006	Lead conversion	Convert an eligible lead and verify customer creation and lead state.
TC-07	CRM-F-007, F-008	Customer profile / assignment	Update a customer profile, assign to Sales Staff and verify correct visibility.
TC-08	CRM-F-009	Interaction tracking	Create and view an interaction linked to the intended customer.
TC-09	CRM-F-010	Follow-up management	Create/update a follow-up and verify its schedule and status.
TC-10	CRM-F-011	Feedback management	Create/update feedback and verify the saved record.
TC-11	CRM-F-012	Retention management	Create/update retention activity and verify the saved state.
TC-12	CRM-F-013	Sales management	Create/update a sale and verify amount and payment status.
TC-13	CRM-F-014	Campaign management	Verify the implemented campaign operations and saved campaign data.
TC-14	CRM-F-015, NF-001	Dashboard / Reports	Verify report values and measure dashboard/list loading time.
TC-15	CRM-F-016	Audit logging	Perform relevant actions and verify expected audit events are recorded.
TC-16	CRM-F-017	GitHub Actions / PR	Verify configured workflow and pull-request checks execute and report status.
TC-17	CRM-F-018	System testing	Execute a core end-to-end workflow across authentication, CRM records and reports.
TC-18	CRM-F-018, F-019	Regression / documentation	Re-run core features after fixes and review document-to-project consistency.
TC-19	CRM-NF-001	Performance	Measure dashboard and list loading with test conditions recorded.
TC-20	CRM-NF-002	Persistent storage	Save/update records, refresh/restart and verify records remain available.
TC-21	CRM-NF-003	Modular structure	Review frontend/backend/data organization for modularity and maintainability.
TC-22	CRM-NF-004, SR-003, SR-004	Safe errors / secret protection	Submit invalid requests and inspect responses/logs for secret disclosure.
TC-23	CRM-SR-005	Record integrity	Attempt unauthorized record modifications and verify rejection.

14. Test Metrics and Reporting

- Number and percentage of planned test cases executed.
- Pass, fail, blocked and not-run counts.
- Defect count, severity and retest outcomes.
- Coverage of functional, non-functional and security requirements.
- Measured dashboard/list loading times and test conditions.

15. Approvals

Role	Name
Project / Test Coordinator	Shweta
Team Member	Avaneesh N
Team Member	Vishwas V Angadi
Team Member	Krishna Pai
Instructor / Reviewer	To be completed